Hints: Linear algebra – 5. Bilinear and quadratic forms

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Use Gram–Schmidt, i.e. $A = QR$ with Q orthogonal and R upper triangular. Then $|\det A| = \prod r_{jj}$, and r_{jj} is the distance from A_j to the span of $A_1, \dots, A_{j-1}$.

Hint 2. That distance is $\leqslant \|A_j\|$, with equality iff A_j is orthogonal to the earlier columns (or the earlier columns are dependent, in which case $\det A = 0$). For $A \succeq 0$ write $A = B^T B$. Then $a_{ii} = \|B_i\|^2$ and $\det A = (\det B)^2$.

Problem 2

(a)

Hint 1. A non-scalar 2×2 matrix has centraliser $\{xE + yA_0\}$. So every $A_k = x_k E + y_k A_0$ is a polynomial in A_0, and so is $\sum A_k^2$.

Hint 2. Let α, β be the eigenvalues of A_0. The eigenvalues of $\sum A_k^2$ are $\sum \lambda_k^2$ and $\sum \mu_k^2$, where $\lambda_k = x_k + y_k\alpha$ and $\mu_k = x_k + y_k\beta$. If α, β are real, both sums are $\geqslant 0$. If not, $\mu_k = \overline{\lambda_k}$ and the determinant is $|\sum \lambda_k^2|^2$.

(b)

Hint 1. Real case: the determinant vanishes only if, say, all $\lambda_k = 0$. Show that then all A_k are multiples of one fixed matrix: use a basis diagonalising A_0, or $A_0 = \alpha E + N$ in the Jordan case. This contradicts $A_2 \neq aA_1$.

Hint 2. Complex case: $\sum \lambda_k^2 = 0$. Then $\sum A_k^2 = p(A_0)$ for a real polynomial p of degree $\leqslant 1$ (reduce using Cayley–Hamilton) vanishing at α and $\bar{\alpha}$. So $p = 0$.

Problem 3

Hint 1. $A = \frac{1}{2}E + K$ with K real skew-symmetric. The eigenvalues of K are 0 and pairs $\pm ib$.

Hint 2. So $\det A = (\frac{1}{2})^{n-2m} \prod_j (\frac{1}{4} + b_j^2) > 0$.

Problem 4

(a)

Hint 1. $A - A^*$ is skew-Hermitian, so $a = it$ is purely imaginary. Write $A = H + aE$ with $H = \frac{1}{2}(A + A^*)$ Hermitian.

Hint 2. The eigenvalues of A are $h_j + it$ with h_j real. Hence $|h_j + it| \geqslant |t| = |a|$.

(b)

Hint 1. Equality forces all $h_j = 0$, so $H = 0$ (Hermitian with zero spectrum) and $A = aE$.

Problem 5

Hint 1. Average over all 2^m sign choices: expand $\operatorname{tr}(\sum \varepsilon_i A_i)^2$.

Hint 2. The cross terms $\varepsilon_i\varepsilon_j \operatorname{tr}(A_iA_j)$, $i \neq j$, average to 0. So the mean equals $\sum \operatorname{tr} A_i^2$, and some choice is at least the mean.

Problem 6

(a)

Hint 1. Take $C = \sqrt{\lambda}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, so $C^2 = -\lambda E$. It remains to write $M = A + \lambda E$ as a square B^2.

Hint 2. For large λ, $\det M > 0$ and $\operatorname{tr} M > 0$. Put $s = \sqrt{\det M}$. By Cayley–Hamilton, $B = (M + sE)/\sqrt{\operatorname{tr} M + 2s}$ satisfies $B^2 = M$.

(b)

Hint 1. If $BC = CB$, then $B^2 + C^2 = (B + iC)(B - iC)$, and $B - iC = \overline{B + iC}$.

Hint 2. So $\det(B^2 + C^2) = |\det(B + iC)|^2 \geqslant 0$, whereas the given matrix has determinant -1.

Problem 7

Hint 1. Block Gaussian elimination is a congruence: $\begin{pmatrix} E & 0 \\ -B^TA^{-1} & E \end{pmatrix}\begin{pmatrix} A & B \\ B^T & C \end{pmatrix}\begin{pmatrix} E & -A^{-1}B \\ 0 & E \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & C - B^TA^{-1}B \end{pmatrix}$.

Hint 2. Hence $\det M = \det A \cdot \det S$ with $S = C - B^TA^{-1}B$, and $0 \preceq S \preceq C$. For $0 \preceq S \preceq C$ with $C \succ 0$, the matrix $C^{-1/2}SC^{-1/2}$ has eigenvalues in $[0, 1]$. This gives $\det S \leqslant \det C$. If C is singular, then S is too, and both sides are 0.

Problem 8

Hint 1. Simultaneously diagonalise (Theorem 3 of the class work): $S^TAS = E$ and $S^TBS = \operatorname{diag}(d_i)$ with $d_i > 0$. Both sides scale by the same power of $\det S$.

Hint 2. The claim reduces to $\prod(t + (1 - t)d_i) \geqslant \prod d_i^{1-t}$, which is weighted AM–GM factor by factor.

Problem 9

Hint 1. If A and B are both singular, there is nothing to prove. Otherwise let $A \succ 0$ and diagonalise simultaneously: $A \to E$, $B \to \operatorname{diag}(d_i)$, $d_i \geqslant 0$. You need $\prod(1 + d_i)^{1/n} \geqslant 1 + \prod d_i^{1/n}$.

Hint 2. Divide by the left side and apply AM–GM to each of the two products: $\prod(\frac{1}{1+d_i})^{1/n} + \prod(\frac{d_i}{1+d_i})^{1/n} \leqslant \frac{1}{n}\sum \frac{1}{1+d_i} + \frac{1}{n}\sum \frac{d_i}{1+d_i} = 1$.

Hint 3. The reduced inequality is exactly Brunn–Minkowski for two axis-parallel boxes with sides 1 and d_i. Generally, $\det^{1/n}$ on the positive semidefinite cone behaves like $\operatorname{vol}^{1/n}$ on convex bodies: n-homogeneous, with a concave n-th root.

Problem 10

Hint 1. Look at inner products instead of distances: $\langle v_i, v_j\rangle = \frac{1}{2}(|v_i|^2 + |v_j|^2 - |v_i - v_j|^2)$ is rational (use $v_0 = 0$).

Hint 2. The Gram matrix G of $v_1, \ldots, v_{n+1}$ is a rational $(n + 1) \times (n + 1)$ matrix of rank $\leqslant n$. So $Gq = 0$ for some nonzero $q \in \mathbb{Q}^{n+1}$.

Hint 3. Then $|\sum q_iv_i|^2 = q^TGq = 0$.